STAFF REVIEW · H446 307 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. It may avoid a slower main-memory access.

2. Accurate mechanism, context and supported consequences.

3. Credit for a correct intermediate step or justified inference.

4. 58_{16}, decimal 88 (64+16+8).

I Do — source worked steps

1. Predict first — Name the mechanism—overlapping stages of different instructions—rather than speeding an individual stage.

2. Check reasoning — State a limitation, such as dependency stalls or branch flushes.

We Do — guided CHECK

1. Predict first
Name the matching attribute and type of join rather than a generic table combination.
Check reasoning
Explain that rows with no matching StudentID in both tables are excluded.

You Do — source mark schemes

1. Separate instruction/data memory pathways [2], potential simultaneous fetch/data access [2], bottleneck reduced in some designs [1], not unconditional speed advantage [1].

2. +6=00000110 [2], invert=11111001 [2], +1=11111010 [2]; initial answer equals −7.

3. Specific access benefit [1], risk of outages [1], security/privacy [1], costs [1], mitigation [1], justified conclusion relative to context [1].

Exit Ticket — source answer

1. False. Working must be shown where the mark scheme requires it.